Wildfire Project — Project Review 4

Building and Evaluating Classification Models for Fire Occurrence

OBJECTIVE

End-to-End Classification Pipeline and Model Comparison

Scope: Daily California Weather & Fire Records (1984–2023)

SESSION CONTEXT

Milestone 4 Submission

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Problem and Dataset

Goal: Classify daily observations into "recorded fire start" vs. "no fire start" using daily weather indicators.

Following technical review feedback, we replaced the small 244-row dataset with the **California Weather and Fire Prediction Dataset**.

The integrated record combines NOAA daily weather observations with incident reports from CAL FIRE.

14,988
TOTAL DAILY RECORDS

1984–2025
TEMPORAL COVERAGE

60x
DATASET ENLARGEMENT

Note: This is retrospective same-day historical classification; same-day weather may be contemporaneous with the incident.

Data Quality and Coverage Decision

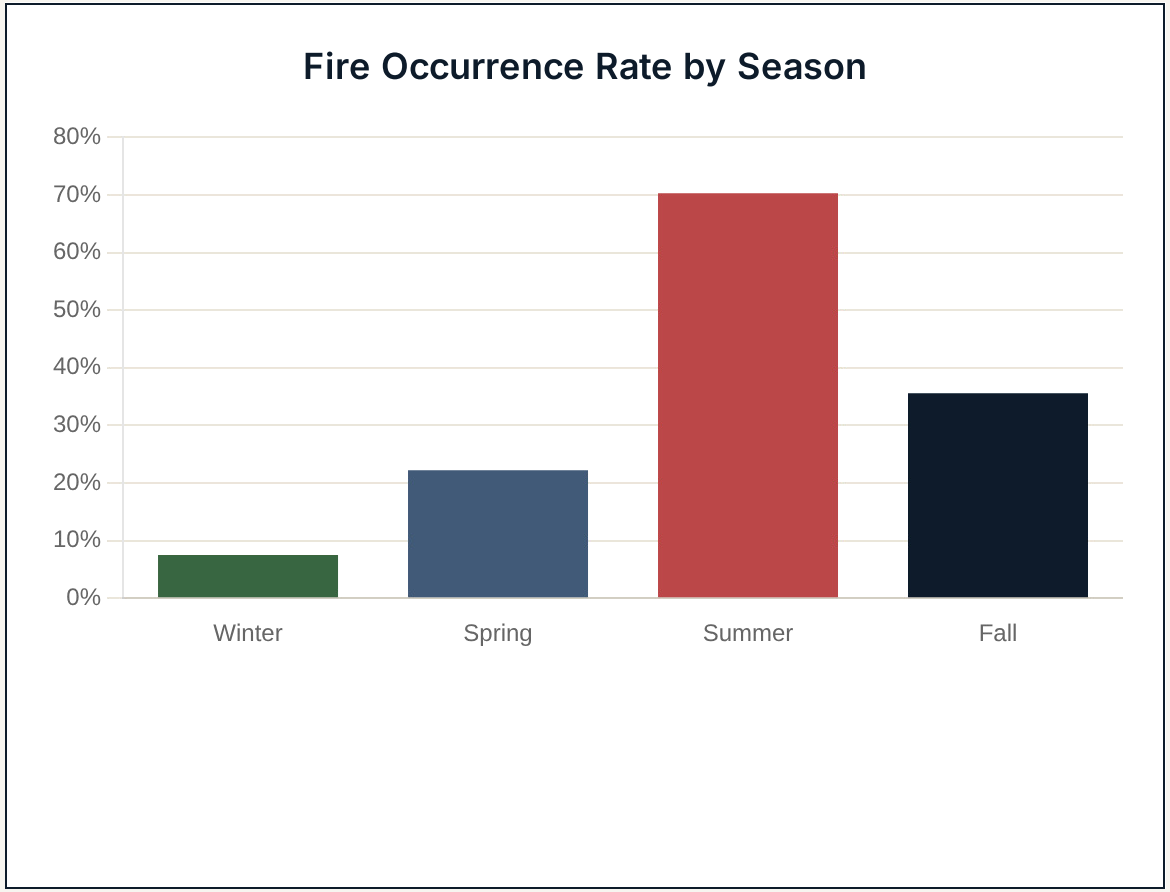
METRIC	COUNT
Total Source Rows	**14,988**
Duplicate Rows	**0**
Missing Wind Observations	**3**
Excluded Rows (2024–2025)	**378**
Final Analysis & Modeling Sample	**14,610**

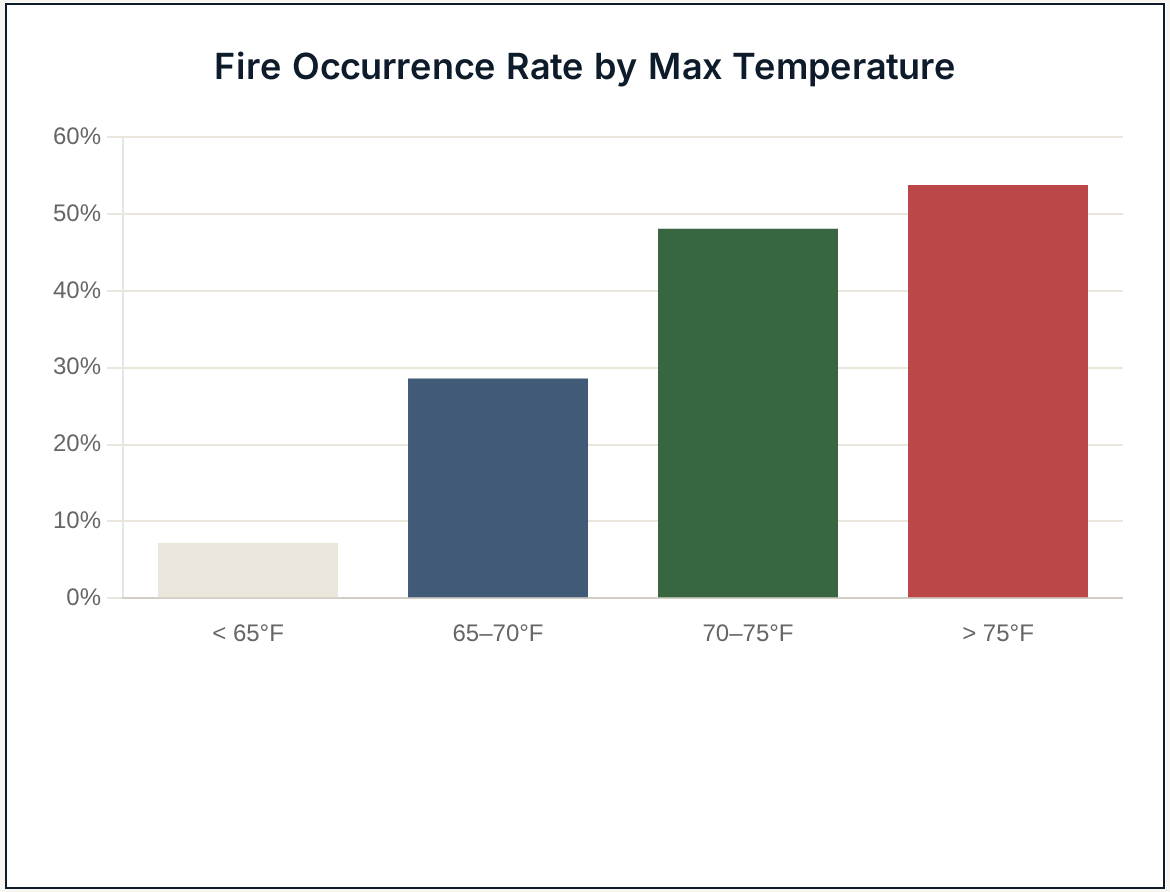
2024–2025 Coverage Audit

The audit revealed that 100% of dates in 2024 and early 2025 were labeled as "no fire start" (False), deviating from the historical record.

To avoid polluting models with potentially incomplete label coverage, we restricted the scope to the validated **1984–2023** period.

What Did We Observe?

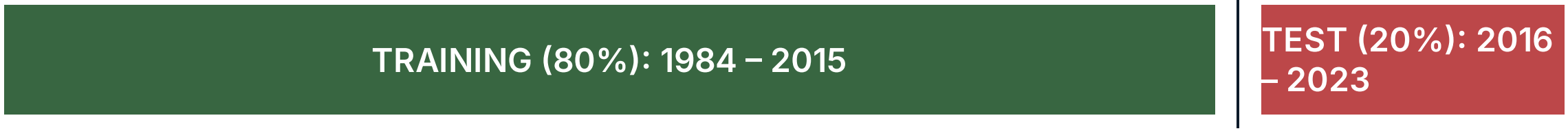




Strong concentration in **Summer (70.3%)** compared to Winter (7.5%). Season acts as a proxy for complex environmental and human factors.

Clear positive correlation with temperature: Days exceeding **75°F** saw fire starts in 53.8% of cases, dropping to 7.2% below 65°F.

Preparation and Chronological Split



Feature Correction

Source "lagged" fields were found to include current-day data. We rebuilt the 7-day rolling precipitation and wind speed using only preceding rows to prevent data leakage.

Pipeline Discipline

Imputation, Scaling, and One-Hot Encoding parameters were fitted on the training set only. Each cross-validation fold followed the same temporal restriction.

Candidate Models and Selection Metric

Logistic Regression

A transparent linear baseline with balanced class weights. Efficient to train and highly interpretable for regulatory or engineering reviews.

Random Forest

A robust ensemble model capable of capturing non-linear feature interactions between seasonal, daily, and rolling weather indicators.

Hist Gradient Boosting

State-of-the-art tree-based learner designed for large-scale tabular data, handling varied distributions and large feature spaces efficiently.

Primary Metric: Average Precision (AP)

AP summarizes the precision-recall trade-off across all thresholds. We selected this over Accuracy to account for the class imbalance and the temporal shift in fire occurrence rates.

Chronological Model Comparison

Average of 5-fold expanding window time-series CV on training data (1984–2015)

MODEL CANDIDATE	MEAN AVERAGE PRECISION	BALANCED ACCURACY	MEAN F1 SCORE
Logistic Regression	**0.652 ± 0.102**	**0.742**	**0.646**
Random Forest	0.648 ± 0.113	0.755	0.660
Hist Gradient Boosting	0.623 ± 0.126	0.698	0.577
Majority Baseline	0.321 ± 0.071	0.500	0.000

Selection Result: Logistic Regression was selected as the winner. While Random Forest shows slightly higher F1, Logistic Regression achieved the highest mean AP with lower structural complexity. The cross-validation score is used for selection and is not a final unbiased estimate.

Final Holdout Test Results

Logistic Regression | Test Period: 2016–2023 (2,922 days) | Threshold = 0.5

71.7%
ACCURACY

72.8%
BALANCED ACCURACY

64.2%
FIRE PRECISION

87.0%
FIRE RECALL

73.8%
F1 SCORE

0.797
AVERAGE PRECISION

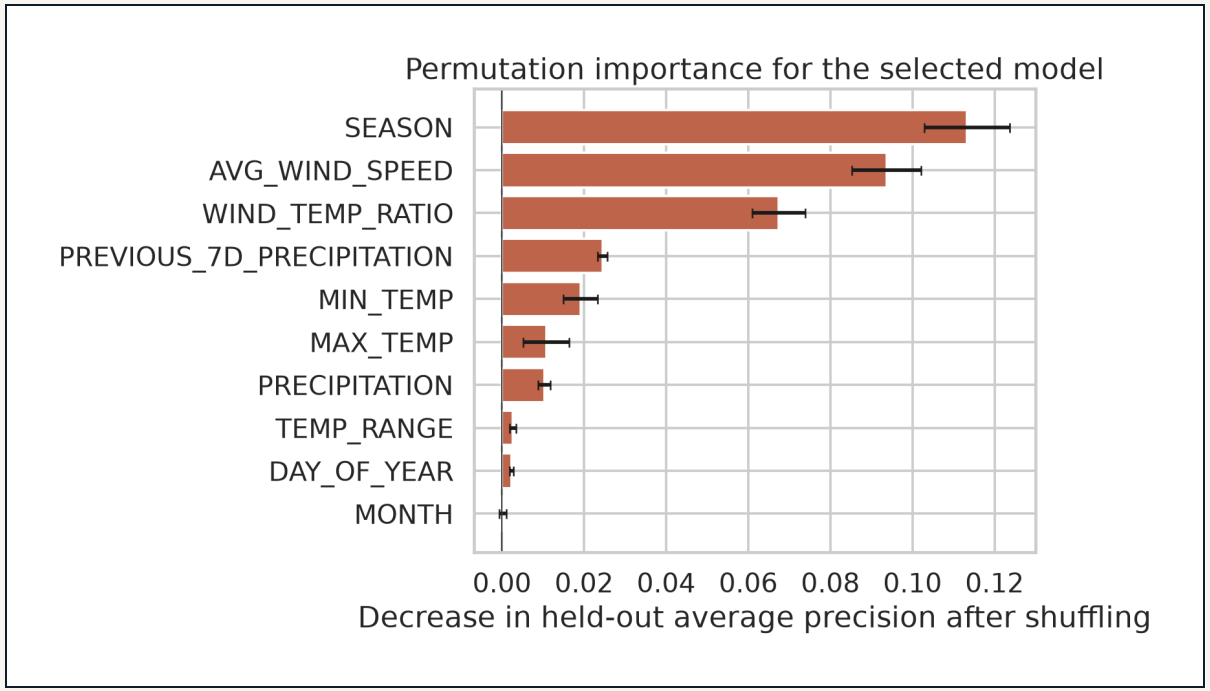
Confusion Matrix (Counts)

	Pred: No Fire	Pred: Fire
Obs: No Fire	925	653
Obs: Fire	175	1,169

High sensitivity (87% recall) ensures most fire starts are captured, but at the cost of significant false alarms (653 days). The default 0.5 threshold is not optimized for specific operational response costs.

Interpreting the Results

Permutation Importance: Decrease in Average Precision when shuffling features on holdout



Seasonal Dominance

SEASON was the strongest predictor for ranking fire starts. This confirms that broad climatic cycles drive the base likelihood in the dataset.

Wind and Ratio Signal

Average Wind Speed and the Wind-Temp Ratio followed as key indicators, aligning with the engineering intuition that wind drives fire behavior.

Diagnostic, Not Causal

Importance metrics describe model dependencies in this sample. They do not prove that changing a feature value causes a fire start.

Disclaimer: Classifier scores are not calibrated risk probabilities. The analysis is limited to retrospective classification on same-day weather records.

Limitations and Next Steps

Critical Constraints

- → **Temporal Shift:** Positive fire start rates rose from 31% (train) to 46% (test), limiting direct generalization.
- → **Same-Day Features:** Using weather recorded on the same day as the fire start precludes claims of early warning capability.
- → **Label Integrity:** The 2024–2025 all-negative pattern suggests potential target coverage gaps in recent records.

Future Directions

- → **Forecasting Horizon:** Shift inputs by -24h to evaluate 1-day-ahead predictive performance using only prior information.
- → **Spatial Features:** Integrate fuel moisture levels, vegetation indices, and regional topography to improve precision.
- → **Threshold Tuning:** Calibrate decision cutoffs against a validation set based on firefighting resource costs.

Data-Driven Wildfire Classification

We successfully scaled the project from a 244-row pilot to a robust 14,000+ row daily analysis. By applying rigorous chronological splitting and feature correction, we established a high-sensitivity baseline (87% Recall) using Logistic Regression.

This project provides the foundation for an evidence-based approach to wildfire monitoring, identifying Season and Wind as critical same-day indicators while establishing clear methodological guardrails for future predictive systems.

Questions & Answers

Thank you for your time and attention.

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